

**7.0.1 NIELIT 2016 MAR Scientist C - Section C: 62**

The excess 3 code is also called

- A. cyclic redundancy code
- B. weighted code
- C. self complementing code
- D. algebraic code

nielit2016mar-scientistc digital-logic

Answer key

7.0.2 NIELIT 2016 MAR Scientist C - Section C: 36

Odd parity of word can be conveniently tested by

- A. OR gate
- B. AND gate
- C. NOR gate
- D. XOR gate

nielit2016mar-scientistc digital-logic

Answer key

7.1 Adder (2)**7.1.1 Adder: NIELIT 2016 DEC Scientist B (IT) - Section B: 33**

How many inputs are required in Full Adder Circuit?

- A. 2
- B. 3
- C. More than two inputs
- D. None of the above

nielit2016dec-scientistb-it digital-logic combinational-circuit adder

Answer key

7.1.2 Adder: NIELIT 2021 Dec Scientist A - Section B: 94

The number of full and half-adders required to add 16-bit numbers is :

- A. 8 half-adders, 8 full-adders
- B. 1 half-adders, 15 full-adders
- C. 16 half-adders, 0 full-adders
- D. 4 half-adders, 12 full-adders

nielit2021dec-scientista digital-circuits adder

Answer key

7.2 Binary Arithmetic (1)**7.2.1 Binary Arithmetic: NIELIT 2021 Dec Scientist B - Section B: 81**

The two numbers represented in Signed 2^i 's complement form are

$A = 11101101$ and $B = 11100110$. If B is subtracted from A , the value obtained in signed 2^i 's complement is :

- A. 111000101 B. 00000111 C. 11111000 D. 10000011

nielit2021dec-scientistb number-representation twos-complement binary-arithmetic data-structures

Answer key 

7.3 Bit Representation (1)

7.3.1 Bit Representation: NIELIT 2021 Dec Scientist B - Section B: 4



Maximum number of bits required to represent any character from ASCII code set is:

- A. 10 B. 8 C. 7 D. 3

nielit-2021-it-dec-scientistb number-representation ascii bit-representation

Answer key 

7.4 Bitwise Operations (1)

7.4.1 Bitwise Operations: NIELIT 2021 Dec Scientist A - Section B: 113



What happens when a bit string is XORed with itself n times as shown below ?

$[B \oplus (B \oplus (B \oplus (B \dots n \text{ times})$

- A. Complements when n is even B. Complements when n is odd
C. Divides by 2^n always D. Remains unchanged when n is even

nielit2021dec-scientista bitwise-operations logical-reasoning

Answer key 

7.5 Boolean Algebra (16)

7.5.1 Boolean Algebra: NIELIT 2016 MAR Scientist B - Section C: 1



Which of the following logic expression is incorrect?

- A. $1 \oplus 0 = 1$ B. $1 \oplus 1 \oplus 0 = 1$
C. $1 \oplus 1 \oplus 1 = 1$ D. $1 \oplus 1 = 0$

nielit2016mar-scientistb digital-logic boolean-algebra

Answer key 

7.5.2 Boolean Algebra: NIELIT 2017 DEC Scientific Assistant A - Section B: 55

In Boolean algebra $1 + 1 + 1 + 1 \dots 800$ times ones = _____ .



- A. 1 B. 0 C. 11 D. 800

nielit2017dec-assistanta digital-logic boolean-algebra

Answer key 

7.5.3 Boolean Algebra: NIELIT 2017 July Scientist B (CS) - Section B: 18



If $A \oplus B = C$, then which one of the following is true?

- A. $A \oplus C = B$
- B. $B \oplus C = A$
- C. $A \oplus B \oplus C = 0$
- D. Both (A) and (B)

nielit2017july-scientistb-cs digital-logic boolean-algebra

Answer key

7.5.4 Boolean Algebra: NIELIT 2017 July Scientist B (IT) - Section B: 51



In digital logic, if $A \oplus B = C$, then which one of the following is true?

- A. $A \oplus C = B$
- B. $B \oplus C = A$
- C. $A \oplus B \oplus C = 0$
- D. Both (A) and (B)

nielit2017july-scientistb-it digital-logic boolean-algebra

Answer key

7.5.5 Boolean Algebra: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 20



The possible number of Boolean function of 3 variables X, Y and Z such that $f(X, Y, Z) = f(X', Y', Z')$

- A. 8
- B. 16
- C. 64
- D. 32

nielit2017oct-assistanta-cs digital-logic boolean-algebra

Answer key

7.5.6 Boolean Algebra: NIELIT 2018-59



$(A + C')(B' + C')$ simplifies to

- A. $AC' + B'$
- B. $C(A' + B')$
- C. $BC' + A$
- D. $AB' + C'$

nielit-2018 digital-logic boolean-algebra

Answer key

7.5.7 Boolean Algebra: NIELIT 2018-66



_____ number of gates are required to implement the Boolean function $(AB + C)$ with using only 2-input NOR gates.

- A. 2
- B. 3
- C. 4
- D. 5

nielit-2018 digital-logic boolean-algebra

Answer key

7.5.8 Boolean Algebra: NIELIT 2018-71



In the given truth table, $f(x, y)$ represent the Boolean function.

x	y	$f(x, y)$
0	0	1
0	1	0
1	0	0
1	1	1

A. $x \leftrightarrow y$

B. $x \wedge y$

C. $x \vee y$

D. $x \rightarrow y$

nielit-2018 boolean-algebra

Answer key

7.5.9 Boolean Algebra: NIELIT 2021 Dec Scientist A - Section B: 106



Let $f(A, B) = \bar{A} + B$, Simplified expression for function $f(f(x+y, y), z)$ is :

A. $\bar{x} + z$

B. xyz

C. $x\bar{y} + z$

D. None of the options

nielit2021dec-scientista boolean-algebra digital-logic

Answer key

7.5.10 Boolean Algebra: NIELIT 2021 Dec Scientist A - Section B: 52



Which of the following Boolean algebra rules is correct ?

A. $A \cdot \bar{A} = 1$

B. $A + AB = A + B$

C. $A(A + B) = B$

D. $A + \bar{A}B = A + B$

nielit2021dec-scientista boolean-algebra logical-reasoning digital-logic

Answer key

7.5.11 Boolean Algebra: NIELIT 2021 Dec Scientist B - Section B: 101



Which of the following Boolean expressions is not logically equivalent to all of the rest ?

A. $ab + (cd)' + cd + bd'$

B. $a(b+c) + cd$

C. $ab + ac + (cd)'$

D. $bd' + c'd' + ab + cd$

nielit2021dec-scientistb boolean-algebra logical-reasoning digital-logic

7.5.12 Boolean Algebra: NIELIT 2021 Dec Scientist B - Section B: 114



If data stored in $AC = 5F$ h and $DR = C2$ h what is value of AC after $AC \wedge DR$ operation ?

A. 9D

B. 42

C. DF

D. DE

nielit2021dec-scientistb digital-logic boolean-algebra

Answer key

7.5.13 Boolean Algebra: NIELIT 2021 Dec Scientist B - Section B: 50



When factorizing the Boolean equation $Y = A\bar{B} + AB$, the result will be :

- A. $A\bar{B}$ B. AB C. A D. B

nielit2021dec-scientistb boolean-algebra digital-logic

Answer key

7.5.14 Boolean Algebra: NIELIT 2021 Dec Scientist B - Section B: 57



The complement of the expression $Y = ABC + AB\bar{C} + \bar{A}\bar{B}C + \bar{A}BC$ is :

- A. $(\bar{A} + \bar{B})(A + \bar{C})$
B. $(\bar{A} + B)(A + C)$
C. $(A + \bar{B})(\bar{A} + C)$
D. $(A + \bar{B})(A + \bar{C})$

nielit-2021-it-dec-scientistb digital-logic boolean-algebra

7.5.15 Boolean Algebra: NIELIT 2021 Dec Scientist B - Section B: 94



The complement of the expression $Y = ABC + AB\bar{C} + \bar{A}\bar{B}C + \bar{A}BC$ is :

- A. $(\bar{A} + \bar{B})(A + \bar{C})$
B. $(\bar{A} + B)(A + C)$
C. $(A + \bar{B})(\bar{A} + C)$
D. $(A + \bar{B})(A + \bar{C})$

nielit2021dec-scientistb digital-logic boolean-algebra

Answer key

7.5.16 Boolean Algebra: NIELIT Scientist B 2020 November: 97



If x, y, z are Boolean variable then $(x + \bar{y})(x \cdot \bar{y} + x \cdot z)(\bar{x} \cdot \bar{z} + \bar{y})$ is equal to :

- A. $x \cdot \bar{y}$ B. $x \cdot \bar{y} + z$
C. $x \cdot \bar{z}$ D. none of the options

nielit-scb-2020 digital-logic boolean-algebra

Answer key

7.6.1 Booths Algorithm: NIELIT 2016 DEC Scientist B (IT) - Section B: 58



How many Addition and Subtraction are required if you perform multiplication of 5 (Multiplicand) and -30 (Multiplier) using Booth Algorithm?

- A. 2,1 B. 1,2 C. 1,1 D. 2,2

nielit2016dec-scientistb-it digital-logic booths-algorithm

Answer key

7.7 CO and Architecture (1)

7.7.1 CO and Architecture: NIELIT 2021 Dec Scientist B - Section B: 5



ALU does not perform the operation of:

- A. Data transfer B. Logical operation
C. Arithmetic operation D. Comparison operation

nielit-2021-it-dec-scientistb co-and-architecture

Answer key

7.8 Carry Generator (1)

7.8.1 Carry Generator: NIELIT 2016 MAR Scientist B - Section C: 2



In which of the following adder circuits, the carry look ripple delay is eliminated?

- A. Half adder B. Full adder
C. Parallel adder D. Carry-look ahead adder

nielit2016mar-scientistb digital-logic combinational-circuit adder carry-generator

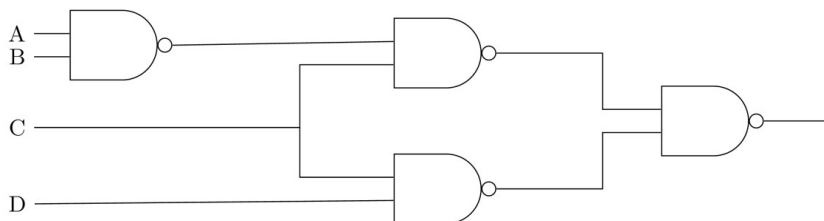
Answer key

7.9 Circuit Output (1)

7.9.1 Circuit Output: NIELIT 2021 Dec Scientist B - Section B: 73



The logic expression for the output of following circuit is:



- A. $\overline{A}C + \overline{B}C + CD$ B. $A\overline{C} + B\overline{C} + \overline{C}D$
C. $ABC + \overline{C}D$ D. $\overline{A}B + \overline{B}C + \overline{C}D$

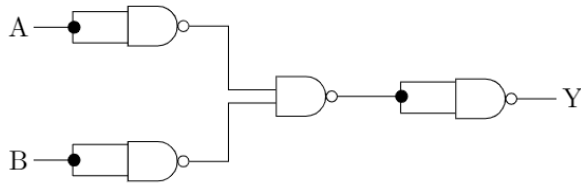
nielit-2021-it-dec-scientistb digital-circuits circuit-output nielit2021dec-scientistb

Answer key

7.10

Combinational Circuit (11)

7.10.1 Combinational Circuit: NIELIT 2016 DEC Scientist B (CS) - Section B: 4



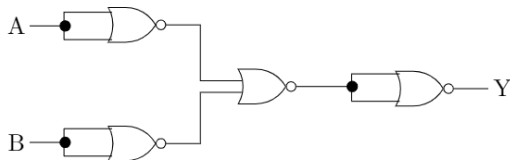
The Circuit is equivalent to:

- A. *OR* Gate B. *NOR* Gate C. *AND* Gate D. *EX – OR* Gate

nielit2016dec-scientistb-cs digital-logic combinational-circuit

Answer key

7.10.2 Combinational Circuit: NIELIT 2016 DEC Scientist B (IT) - Section B: 4



The Circuit is equivalent to:

- A. EX-OR Gate B. NAND Gate C. OR Gate D. AND Gate

nielit2016dec-scientistb-it digital-logic combinational-circuit

Answer key

7.10.3 Combinational Circuit: NIELIT 2016 MAR Scientist B - Section C: 10



Disadvantage of dynamic RAM over static RAM is

- A. higher power consumption.
B. variable speed.
C. need to refresh the capacitor charge every once in two milliseconds.
D. higher bit density.

nielit2016mar-scientistb digital-logic combinational-circuit

Answer key

7.10.4 Combinational Circuit: NIELIT 2016 MAR Scientist B - Section C: 7



How many RAM chips of size $(256K \times 1 \text{ bit})$ are required to build 1M Byte memory?

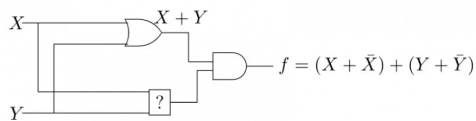
- A. 8 B. 10 C. 24 D. 32

nielit2016mar-scientistb digital-logic combinational-circuit

Answer key

7.10.5 Combinational Circuit: NIELIT 2017 July Scientist B (CS) - Section B: 19

To make the following circuit a tautology '?' marked box should be



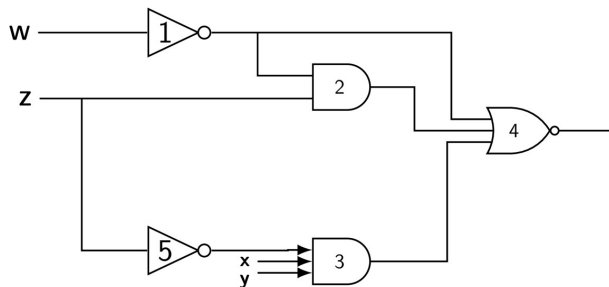
- A. OR gate B. AND gate C. NAND gate D. EX-OR gate

nielit2017july-scientistb-cs digital-logic combinational-circuit

Answer key

7.10.6 Combinational Circuit: NIELIT 2017 July Scientist B (CS) - Section B: 20

In the following gate network which gate is redundant?



- A. Gate no.1 B. Gate no.2 C. Gate no.3 D. Gate no.4

nielit2017july-scientistb-cs digital-logic combinational-circuit

Answer key

7.10.7 Combinational Circuit: NIELIT 2017 July Scientist B (CS) - Section B: 21

The combinational circuit given below is implemented with two NAND gates. To which of the following individual gates is its equivalent?



- A. NOT B. OR C. AND D. XOR

nielit2017july-scientistb-cs digital-logic combinational-circuit easy

Answer key

7.10.8 Combinational Circuit: NIELIT 2017 July Scientist B (IT) - Section B: 49

Which one of the following is true?



- A. NAND gate and AND gate both are universal gates
B. NOR gate and OR gate both are universal gates

- C. NAND gate and OR gate both are universal gates
D. NAND gate and NOR gate both are universal gates

nielit2017july-scientistb-it digital-logic combinational-circuit easy

Answer key

7.10.9 Combinational Circuit: NIELIT 2018-73



Minimum ____ full adders and ____ half adders are required by the BCD adder to add two decimal digits.

- A. 3,2 B. 9,4 C. 6,5 D. 5,2

nielit-2018 digital-logic combinational-circuit

Answer key

7.10.10 Combinational Circuit: NIELIT Scientific Assistant A 2020 November: 111



Encodes are made by three _____ gates.

- A. AND B. OR C. NAND D. XOR

nielit-sta-2020 digital-logic combinational-circuit

Answer key

7.10.11 Combinational Circuit: NIELIT Scientific Assistant A 2020 November: 54



How many AND, OR and XOR gates are required for implementation of full adder?

- A. 1,2,2 B. 2,2,1 C. 3,2,2 D. 3,0,1

nielit-sta-2020 digital-logic combinational-circuit

Answer key

7.11 Communication (1)

7.11.1 Communication: NIELIT 2022 April Scientist B | Section B | Question: 74



In context of analog & digital communication, as the length of a long-wire antenna is increased, _____

- A. the number of lobes also increase B. the number of lobes decrease
C. efficiency increase D. the number of nodes decreases

nielit2022apr-scientistb communication

Answer key

7.12 Decoder (1)

7.12.1 Decoder: NIELIT 2018-39



A RAM chip has a capacity of 1024 words of 8 bits each ($1K \times 8$). The number of 2×4 decoders with enable line needed to construct a $32K \times 8$ RAM from $1K \times 8$ RAM is

- A. 4 B. 5 C. 6 D. 7

nielit-2018 digital-logic combinational-circuit decoder

Answer key

7.13

Digital Circuits (4)

7.13.1 Digital Circuits: NIELIT 2021 Dec Scientist A - Section B: 118



In VCO the output frequency is a linear function of its input :

- A. Frequency B. Voltage C. Time period D. None of the option

nielit2021dec-scientista digital-circuits

7.13.2 Digital Circuits: NIELIT 2021 Dec Scientist A - Section B: 56



A demultiplexer is used to :

- A. Route the data from single input to one of many outputs
B. Perform serial to parallel conversion
C. Both (A) and (B)
D. Select data from several inputs and route it to single output

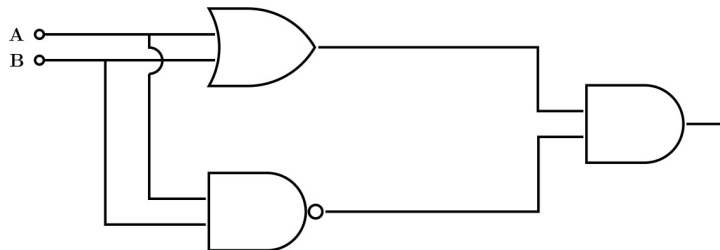
nielit2021dec-scientista digital-circuits combinational-circuit

Answer key

7.13.3 Digital Circuits: NIELIT 2021 Dec Scientist B - Section B: 47



The following circuit depicts the implementation of _____



- A. XOR Gate B. AND Gate C. OR Gate D. NAND Gate

nielit2021dec-scientistb digital-logic digital-circuits combinational-circuit easy

Answer key

7.13.4 Digital Circuits: NIELIT Scientist B 2020 November: 75



What is the basis of KVL ?

- A. Conservation of charge
- B. Conservation of energy
- C. Conservation of power
- D. All of the options

nielit-scb-2020 digital-circuits

Answer key

7.14

Digital Counter (3)

7.14.1 Digital Counter: NIELIT 2017 OCT Scientific Assistant A (CS) - Section D: 10

A 4 bit ripple counter and a 4 bit synchronous counter are made using flip-flops having a propagation delay of 10 ns each. If the worst case delay in the ripple counter and the synchronous counter be R and S respectively, then



- A. $R = 10$ ns, $S = 40$ ns
- B. $R = 40$ ns, $S = 10$ ns
- C. $R = 10$ ns, $S = 30$ ns
- D. $R = 30$ ns, $S = 10$ ns

nielit2017oct-assistanta-cs digital-logic sequential-circuit flip-flop digital-counter

Answer key

7.14.2 Digital Counter: NIELIT 2017 OCT Scientific Assistant A (IT) - Section D: 10

A 4 bit ripple counter and a 4 bit synchronous counter are made using flip-flops having a propagation delay of 10 ns each. If the worst case delay in the ripple counter and the synchronous counter be R and S respectively, then



- A. $R = 10$ ns, $S = 40$ ns
- B. $R = 40$ ns, $S = 10$ ns
- C. $R = 10$ ns, $S = 30$ ns
- D. $R = 30$ ns, $S = 10$ ns

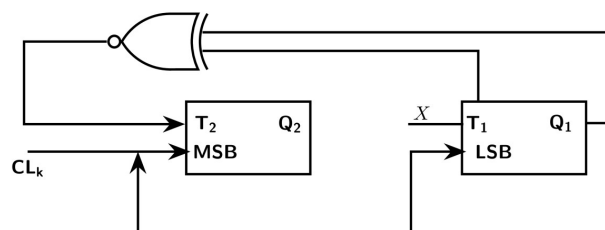
nielit2017oct-assistanta-it digital-logic sequential-circuit flip-flop digital-counter

Answer key

7.14.3 Digital Counter: NIELIT 2021 Dec Scientist B - Section B: 2



Consider the following 2 bit counter using T flip-flops following $0 - 2 - 3 - 1 - 0$ sequence. What should be the value of x ?



- A. Q_2
- B. $Q_1 + Q_2$
- C. $Q_1 \oplus Q_2$
- D. $Q_1 \oplus \overline{Q_2}$

7.15

Error Detection (1)

7.15.1 Error Detection: NIELIT 2021 Dec Scientist A - Section B: 77



A binary sequence $b[n]$ is given as shown below

$$b[n] = \{0, 1, 1, 0, 0, 0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1, 0, 0, 0, 0, 1, 1, 0, 1\}$$

Consider the following statements regarding the above coded sequence :

- It has a DC null in the PSD.
- It possesses error detecting capability.
- It possesses error correcting capability.
- It facilitates clock recovery at the receiver.

Which of the above statements are true?

- A. i, iii, and iv B. i, ii, and iv C. i, ii, and iii D. ii, iii, and iv

nielit2021dec-scientista digital-logic computer-networks error-detection

7.16

Flip Flop (12)

7.16.1 Flip Flop: NIELIT 2016 DEC Scientist B (IT) - Section B: 35



What will be the final output of D flip-Flop if the input string is 0010011100?

- A. 1 B. 0 C. Don't Care D. None of the above

nielit2016dec-scientistb-it digital-logic sequential-circuit flip-flop

Answer key

7.16.2 Flip Flop: NIELIT 2016 MAR Scientist B - Section C: 4



In a ripple counter using edge triggered JK flip-flops, the pulse input is applied to the

- A. clock input of all flip-flops B. clock input of one flip-flop
C. J and K inputs of all flip-flops D. J and K inputs of one flip flop

nielit2016mar-scientistb digital-logic sequential-circuit flip-flop

Answer key

7.16.3 Flip Flop: NIELIT 2016 MAR Scientist C - Section C: 40



If the input J is connected through K input of $J-K$, then flip-flop will behave as a

- A. D type flip-flop B. T type flip-flop
C. S-R flip-flop D. Toggle switch

nielit2016mar-scientistc digital-logic flip-flop

Answer key

7.16.4 Flip Flop: NIELIT 2016 MAR Scientist C - Section C: 41



To build a mod-19 counter the number of flip-flop required is

- A. 3 B. 5 C. 7 D. 8

nielit2016mar-scientistc digital-logic flip-flop

Answer key

7.16.5 Flip Flop: NIELIT 2016 MAR Scientist C - Section C: 44



Which of the following conditions must be met to avoid race around problem?

- A. $\Delta t < t_p < T$ B. $T > \Delta t > t_p$
C. $2t_p < \Delta t < T$ D. none of these

nielit2016mar-scientistc digital-logic flip-flop

Answer key

7.16.6 Flip Flop: NIELIT 2016 MAR Scientist C - Section C: 64



How many flip-flop are needed to divide the input frequency by 64?

- A. 4 B. 5 C. 6 D. 8

nielit2016mar-scientistc digital-logic flip-flop

Answer key

7.16.7 Flip Flop: NIELIT 2017 OCT Scientific Assistant A (CS) - Section C: 8



In a ripple counter using edge-triggered JK flip-flops, the pulse input is applied to

- A. Clock input of all flip-flops B. J and K input of one flip-flop
C. J and K input of all flip flops D. Clock input of one flip-flop

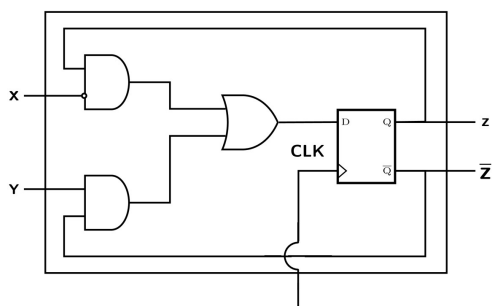
nielit2017oct-assistanta-cs digital-logic sequential-circuit flip-flop

Answer key

7.16.8 Flip Flop: NIELIT 2017 OCT Scientific Assistant A (CS) - Section D: 9



A sequential circuit using D flip-flop and logic gates is shown in Figure, where X and Y are the inputs and Z is the output. The circuit is



- A. S-R Flip-flop with inputs $X = R$ and $Y = S$
 C. J-K Flip-flop with inputs $X = J$ and $Y = K$

- B. S-R Flip-flop with inputs $X = S$ and $Y = R$
 D. J-K Flip-flop with inputs $X = K$ and $Y = J$

nielit2017oct-assistanta-cs digital-logic sequential-circuit flip-flop

Answer key

7.16.9 Flip Flop: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 19



In a ripple counter using edge-triggered JK flip-flops, the pulse input is applied to

- A. Clock input of all flip-flops
 C. J and K input of all flip-flops
 B. J and K input of one flip-flop
 D. Clock input of one flip-flop

nielit2017oct-assistanta-it digital-logic sequential-circuit flip-flop

7.16.10 Flip Flop: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 33



The number of columns in a state table for a sequential circuit with ' m ' flip flops and ' n ' input is

- A. $m + n$
 B. $m + 2n$
 C. $2m + n$
 D. $2m + 2n$

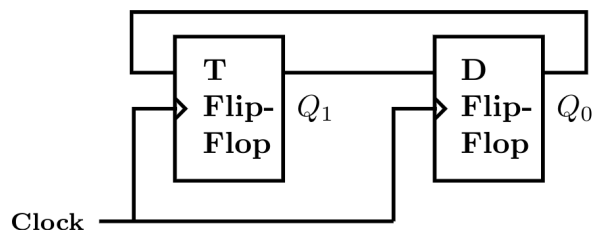
nielit2017oct-assistanta-it digital-logic sequential-circuit flip-flop

Answer key

7.16.11 Flip Flop: NIELIT 2022 April Scientist B | Section B | Question: 94



Consider a combination of T and D flip-flops connected as shown below. The output of the D flipflop is connected to the input of the T flip-flop and the output of the T flip-flop is connected to the input of the D flip-flop.



Initially, both Q_0 and Q_1 are set to 1 (before the 1st clock cycle). The outputs _____

- A. $Q_1 Q_0$ after the 3rd cycle are 11 and after the 4th cycle are 00 respectively
 B. $Q_1 Q_0$ after the 3rd cycle are 11 and after the 4th cycle are 01 respectively
 C. $Q_1 Q_0$ after the 3rd cycle are 00 and after the 4th cycle are 11 respectively
 D. $Q_1 Q_0$ after the 3rd cycle are 01 and after the 4th cycle are 01 respectively

nielit2022apr-scientistb digital-circuits flip-flop sequential-circuit

7.16.12 Flip Flop: NIELIT Scientific Assistant A 2020 November: 85



Which flip-flop is used to make all types of shift registers?

- A. JK flip-flop B. D flip-flop C. T flip-flop D. All the options

nielit-sta-2020 digital-logic sequential-circuit flip-flop

Answer key

7.17

Information Theory (1)

7.17.1 Information Theory: NIELIT 2021 Dec Scientist B - Section B: 103



For a hamming code of parity bit $m = 5$, what is the total bits and data bits ?

- A. (255,247) B. (127,119) C. (31,26) D. (0,8)

nielit2021dec-scientistb error-detection information-theory

Answer key

7.18

K Map (1)

7.18.1 K Map: NIELIT 2016 DEC Scientist B (IT) - Section B: 45



AB \ CD	00	01	11	10
00	X	1		1
01				
11				
10	X	1	1	X

Which will be the equation of simplification of the given K-map?

- A. $AB' + B'CD' + A'B'C'$ B. $AB' + A'B'D' + A'B'C'$
 C. $B'D' + AB' + B'C'$ D. $B'D' + A'B'C' + AB'$

nielit2016dec-scientistb-it digital-logic boolean-algebra k-map

Answer key

7.19

Minimization (1)

7.19.1 Minimization: NIELIT 2016 DEC Scientist B (CS) - Section B: 45



AB \ CD	00	01	11	10
00	X	X		1
01		1		
11		X		
10	X	1		X

What will be the equation of the given K-map?

- A. $A'B'D' + C'D + AB'C'$ B. $B'CD' + AB'C' + A'C'$
 C. $B'D' + C'D$ D. $C'D + B'CD'$

nielit2016dec-scientistb-cs digital-logic k-map minimization

Answer key 

7.20

Multiplexer (3)

7.20.1 Multiplexer: NIELIT 2017 July Scientist B (CS) - Section B: 17



The digital multiplexer is basically a combination logic circuit to perform the operation

- A. AND-AND B. OR-OR C. AND-OR D. OR-AND

nielit2017july-scientistb-cs digital-logic combinational-circuit multiplexer

Answer key 

7.20.2 Multiplexer: NIELIT 2017 July Scientist B (IT) - Section B: 50



Which one of the following is the function of a multiplexer?

- A. To decode information
B. To select 1 out of N input data sources and to transmit it to single channel
C. To transmit data on N lines
D. To perform serial to parallel conversion

nielit2017july-scientistb-it digital-logic multiplexer

Answer key 

7.20.3 Multiplexer: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 30



How many 2-input multiplexers are required to construct a 2^{10} input multiplexer?

- A. 1023 B. 31 C. 10 D. 127

nielit2017oct-assistanta-it digital-logic combinational-circuit multiplexer

Answer key 

7.21

Number Representation (3)

7.21.1 Number Representation: NIELIT 2018-46



The hexadecimal representation of $(632)_8$ is

- A. 19A B. 198 C. 29A D. 291

nielit-2018 digital-logic number-representation easy

Answer key 

7.21.2 Number Representation: NIELIT 2021 Dec Scientist B - Section B: 97



If integer means to 2 bytes of storage then maximum value of a signed integer is :

- A. $2^{16}-1$ B. $2^{15}-1$ C. 2^{16} D. 2^{15}

Answer key

7.21.3 Number Representation: NIELIT Scientist B 2020 November: 120



One-megabyte memory storage in form of bytes is equal to _____

- A. 1024 bytes B. 1024^2 bytes C. 1024^3 bytes D. 1024^4 bytes

Answer key

7.22 Number System (12)

7.22.1 Number System: NIELIT 2016 DEC Scientist B (CS) - Section B: 33



What will be the Excess-3 code for 1001?

- A. 1001 B. 1010 C. 1011 D. 1100

Answer key

7.22.2 Number System: NIELIT 2016 DEC Scientist B (CS) - Section B: 59



The Decimal equivalent of the Hexadecimal number $(A09D)_{16}$ is

- A. 31845 B. 41117 C. 41052 D. 32546

Answer key

7.22.3 Number System: NIELIT 2016 DEC Scientist B (IT) - Section B: 53



What is 2's complement of $(101)_3$?

- A. $(010)_3$ B. $(011)_3$ C. $(121)_3$ D. $(121)_2$

Answer key

7.22.4 Number System: NIELIT 2016 DEC Scientist B (IT) - Section B: 59



The Decimal equivalent of the Hexadecimal number $(AC7B)_{16}$ is:

- A. 32564 B. 44155 C. 50215 D. 43562

Answer key

7.22.5 Number System: NIELIT 2016 MAR Scientist B - Section C: 5



A decimal number has 30 digits. Approximately, how many digits would the binary representation have?

- A. 30 B. 60 C. 90 D. 120

nielit2016mar-scientistb digital-logic number-system

Answer key

7.22.6 Number System: NIELIT 2016 MAR Scientist B - Section C: 6



The result of the subtraction $FD_{16} - 88_{16}$ is

- A. 75_{16} B. 65_{16} C. $5E_{16}$ D. 10_{16}

nielit2016mar-scientistb digital-logic number-system

Answer key

7.22.7 Number System: NIELIT 2017 DEC Scientific Assistant A - Section B: 16



The smallest integer that can be represented by an 8-bit number in 2's complement form is :

- A. -256 B. -128 C. -127 D. 0

nielit2017dec-assistanta digital-logic number-system

Answer key

7.22.8 Number System: NIELIT 2017 DEC Scientific Assistant A - Section B: 26



The number $(25)_6$, in base 6 is equivalent to _____ in binary number system.

- A. 11001 B. 10001 C. 11000 D. 10000

nielit2017dec-assistanta digital-logic number-system

Answer key

7.22.9 Number System: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 1



A decimal has 25 digits. The number of bits needed for its equivalent binary representation is approximately,

- A. 50 B. 74 C. 40 D. 60

nielit2017oct-assistanta-cs digital-logic number-system

Answer key

7.22.10 Number System: NIELIT 2017 OCT Scientific Assistant A (IT) - Section C: 7

A decimal has 25 digits. The number of bits needed for its equivalent binary

representation is approximately



- A. 50 B. 74 C. 40 D. 60

nielit2017oct-assistanta-it digital-logic number-system

Answer key

7.22.11 Number System: NIELIT 2021 Dec Scientist B - Section B: 87



Convert $(503201)_6$ into $(?)_4$

- A. 12122231 B. 11000011 C. 21222301 D. 22323301

nielit2021dec-scientistb number-system

Answer key

7.22.12 Number System: NIELIT 2022 April Scientist B | Section B | Question: 73

Given $\sqrt{(224)_r} = (13)_r$. The value of the radix r is:



- A. 10 B. 8 C. 5 D. 6

nielit2022apr-scientistb number-system quantitative-aptitude

Answer key

7.23

Output (1)

7.23.1 Output: NIELIT 2022 April Scientist B | Section B | Question: 120



Given the function $F = P' + QR$, where F is a function in three Boolean variables P , Q and R and $P' = !P$, consider the following statements.

S1: $F = \sum(4, 5, 6)$

S2: $F = \sum(0, 1, 2, 3, 7)$

S3: $F = \Pi(4, 5, 6)$

S4: $F = \Pi(0, 1, 2, 3, 7)$

Which of the following is true?

- A. (S1)-False, (S2)-True, (S3)-True, (S4)-False
B. (S1)-True, (S2)-False, (S3)-False, (S4)-True
C. (S1)-False, (S2)-False, (S3)-True, (S4)-True
D. (S1)-True, (S2)-True, (S3)-False, (S4)-False

nielit2022apr-scientistb digital-logic boolean-algebra output

7.24

Register (1)

7.24.1 Register: NIELIT 2021 Dec Scientist B - Section B: 111



If the value of Register A = 9B h & Carry = 1. What will be the value of Register A after executing the RORC instruction 1 time ?

- A. AB h B. CD h C. EF h D. AC h

nielit2021dec-scientistb digital-logic co-and-architecture register programming-in-c

Answer key

7.25 Register Allocation (1)

7.25.1 Register Allocation: NIELIT 2016 MAR Scientist C - Section C: 63



The range of the numbers which can be stored in an eight bit register is

- A. -128 to +127 B. -128 to +128
C. -999999 + +999999 D. none of these

nielit2016mar-scientistc digital-logic register-allocation

Answer key

7.26 Sequential Circuit (4)

7.26.1 Sequential Circuit: NIELIT 2016 DEC Scientist B (CS) - Section B: 35



What will be the final output of D Flip-Flop, if the input string is 11010011?

- A. 1 B. 0 C. Don't Care D. None of the above

nielit2016dec-scientistb-cs digital-logic sequential-circuit

Answer key

7.26.2 Sequential Circuit: NIELIT 2016 MAR Scientist B - Section C: 3



The output of a sequential circuit depends on

- A. present inputs only B. past inputs only
C. both present and past inputs D. present outputs only

nielit2016mar-scientistb digital-logic sequential-circuit

Answer key

7.26.3 Sequential Circuit: NIELIT 2016 MAR Scientist C - Section C: 37



A sequential circuit outputs a ONE when an even number (> 0) of one's are input; otherwise the output is ZERO. The minimum number of states required is

- A. 0 B. 1 C. 2 D. 3

nielit2016mar-scientistc digital-logic sequential-circuit

Answer key



If a clock with time period “ T ” is used with n stage shift register, then output of final stage will be delayed by

- A. nT sec B. $(n - 1)T$ sec C. n/T sec D. $(2n - 1)T$ sec

nielit2016mar-scientistc digital-logic sequential-circuit

Answer key

Answer Keys

7.0.1	C	7.0.2	D	7.1.1	B	7.1.2	B	7.2.1	B
7.3.1	B	7.4.1	D	7.5.1	B	7.5.2	A	7.5.3	A;B;C;D
7.5.4	A;B;C;D	7.5.5	B	7.5.6	D	7.5.7	B	7.5.8	A
7.5.9	C	7.5.10	D	7.5.11	A	7.5.12	B	7.5.13	C
7.5.14	A	7.5.15	A	7.5.16	A	7.6.1	B	7.7.1	A
7.8.1	D	7.9.1	A	7.10.1	B	7.10.2	B	7.10.3	B
7.10.4	D	7.10.5	X	7.10.6	B	7.10.7	C	7.10.8	D
7.10.9	D	7.10.10	B	7.10.11	C	7.11.1	A	7.12.1	B
7.13.1	B	7.13.2	C	7.13.3	A	7.13.4	B	7.14.1	B
7.14.2	B	7.14.3	D	7.15.1	TBA	7.16.1	B	7.16.2	B
7.16.3	B	7.16.4	B	7.16.5	B	7.16.6	C	7.16.7	C
7.16.8	D	7.16.9	D	7.16.10	C	7.16.11	B	7.16.12	A
7.17.1	C	7.18.1	C	7.19.1	C	7.20.1	C	7.20.2	B
7.20.3	A	7.21.1	A	7.21.2	B	7.21.3	B	7.22.1	D
7.22.2	B	7.22.3	C	7.22.4	B	7.22.5	C	7.22.6	A
7.22.7	B	7.22.8	B	7.22.9	X	7.22.10	X	7.22.11	C
7.22.12	C	7.23.1	A	7.24.1	B	7.25.1	A	7.26.1	A
7.26.2	C	7.26.3	C	7.26.4	B				